EDS PRACTICAL NO :-3

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PRACTICE ASSIGNMENT

3.1.1 NUMPY ARRAY OPERATIONS

Writea Pythonprogram tocreate aNumPy arraybasedonuserinputanddisplaythefollowing:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of t he array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, and , representing the number of rows and the number of columns.
- The next lines contain space-separated integers representing the elements of the array, entered row by row.

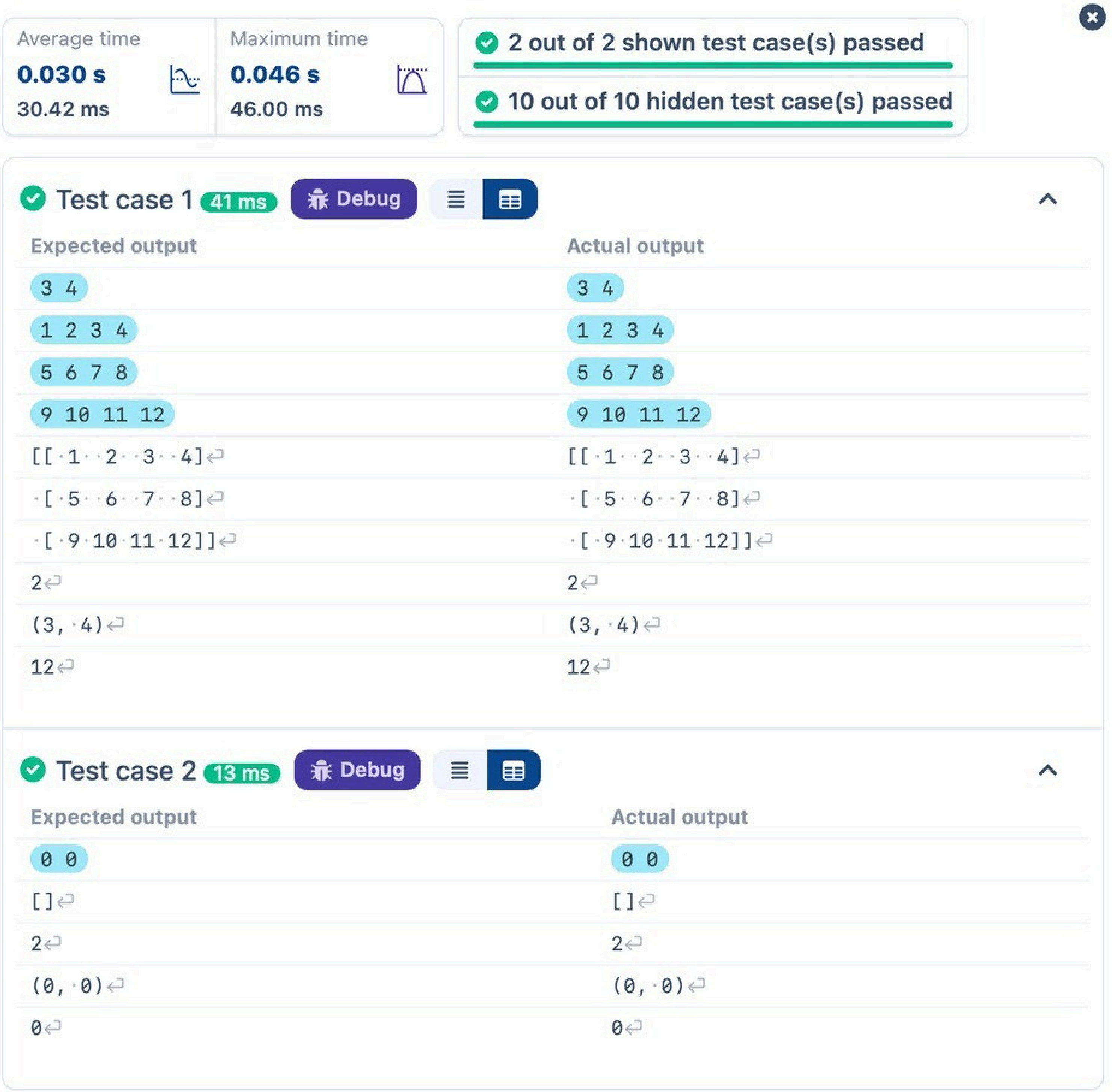
Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

INPUT

```python
import numpy as np

# write your code here...
rows,cols=map(int,input().split())

elements=[]
for _ in range(rows):
    elements.extend(map(int,input().split()))

array=np.array(elements).reshape(rows,cols)

print(array)
print(array.ndim)
print(array.shape)
print(array.size)
```

OUTPUT



SELF ASSIGNMENT

3.2.1 NUMPY MATRIX OPERATIONS

The given code takes two 3x3 matrices, matrix_a, and matrix_b, as input from the user and converts them into NumPy arrays.

Task:
You are required to compute and display the results of the following matrix operations:

1. Addition (matrix_a + matrix_b)
2. Subtraction (matrix_a - matrix_b)
3. Element-wise Multiplication (matrix_a * matrix_b)
4. Matrix Multiplication (matrix_a . matrix_b)
5. Transpose of Matrix A

Input Format:

- The user will input 3 rows for matrix_a , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for matrix_b, each containing 3 integers separated by spaces.

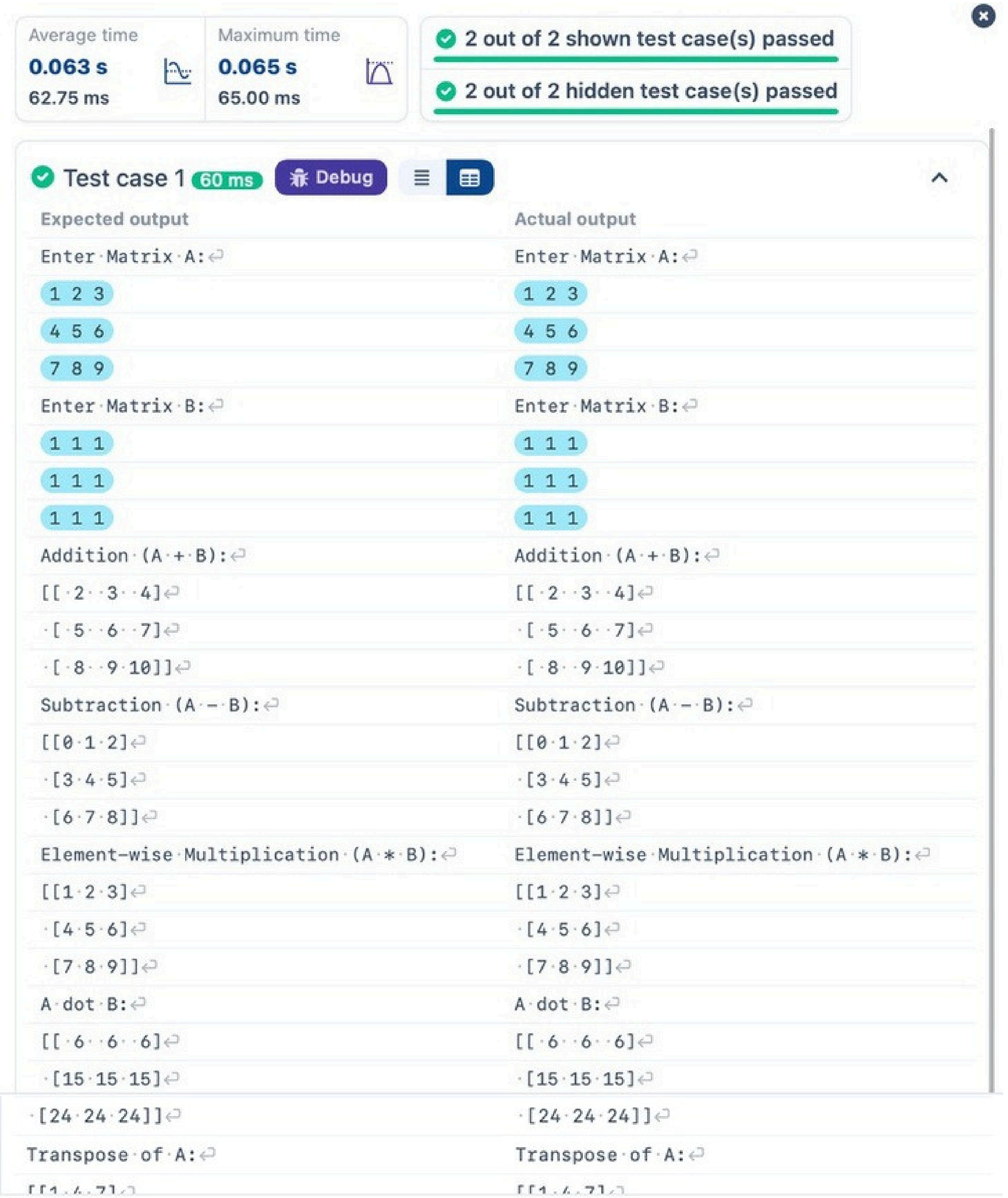
Output Format:
The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

INPUT

```python
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in
range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in
range(3)])


# Addition
addition_result=matrix_a + matrix_b
print("Addition (A + B):")
print(addition_result)
# Subtraction
subtraction_result=matrix_a - matrix_b
print("Subtraction (A - B):")
print(subtraction_result)
# Multiplication (element-wise)
elementwise_multiplication_result=matrix_a * matrix_b
print("Element-wise Multiplication (A * B):")
print(elementwise_multiplication_result)
# Matrix multiplication (dot product)
matrix_multiplcation_result=np.dot(matrix_a,matrix_b)
print("A dot B:")
print(matrix_multiplcation_result)
# Transpose
transpose_a=matrix_a.T
print("Transpose of A:")
print(transpose_a)
```

OUTPUT



3.2.2 NUMPY HORIZONTAL AND VERTICAL STACKING OF ARRAYS

Youaregiventwoarrays arr1and arr2. Youneed toperformhorizontalandverticalstackingoperationsonthem using NumPy.

- Horizontal Stacking: Stack the two matrices horizontally (side by side).
- Vertical Stacking: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

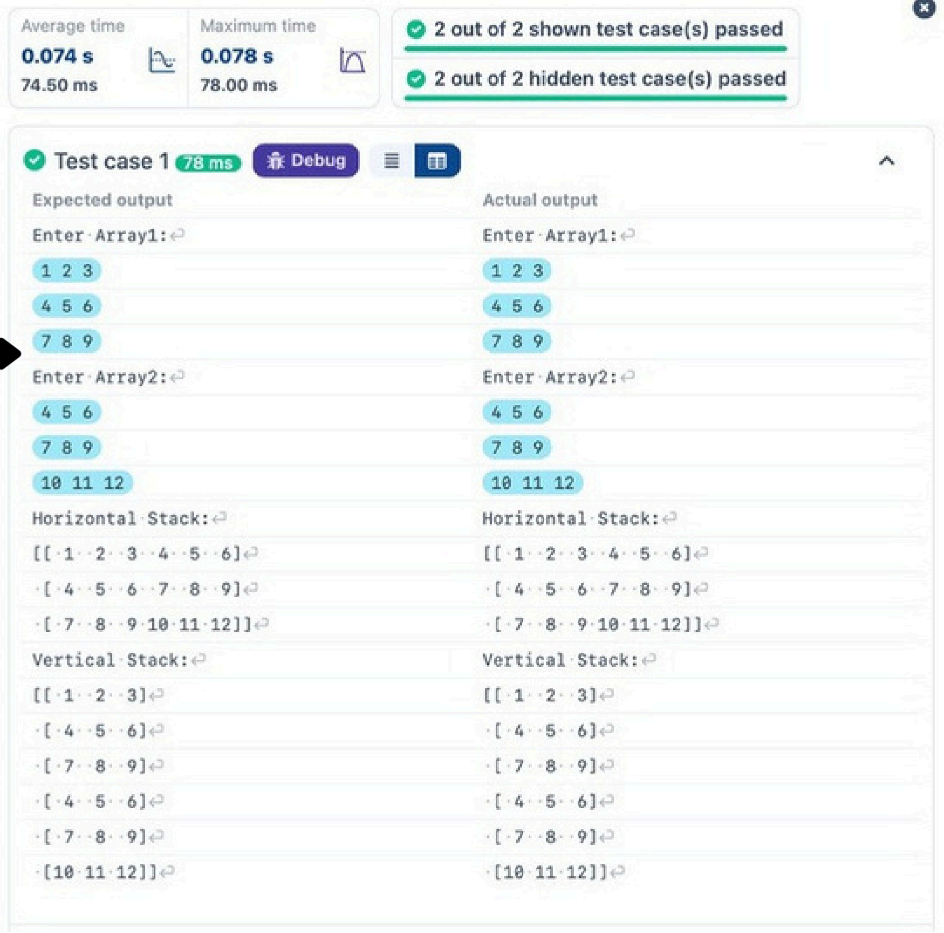
Output Format:

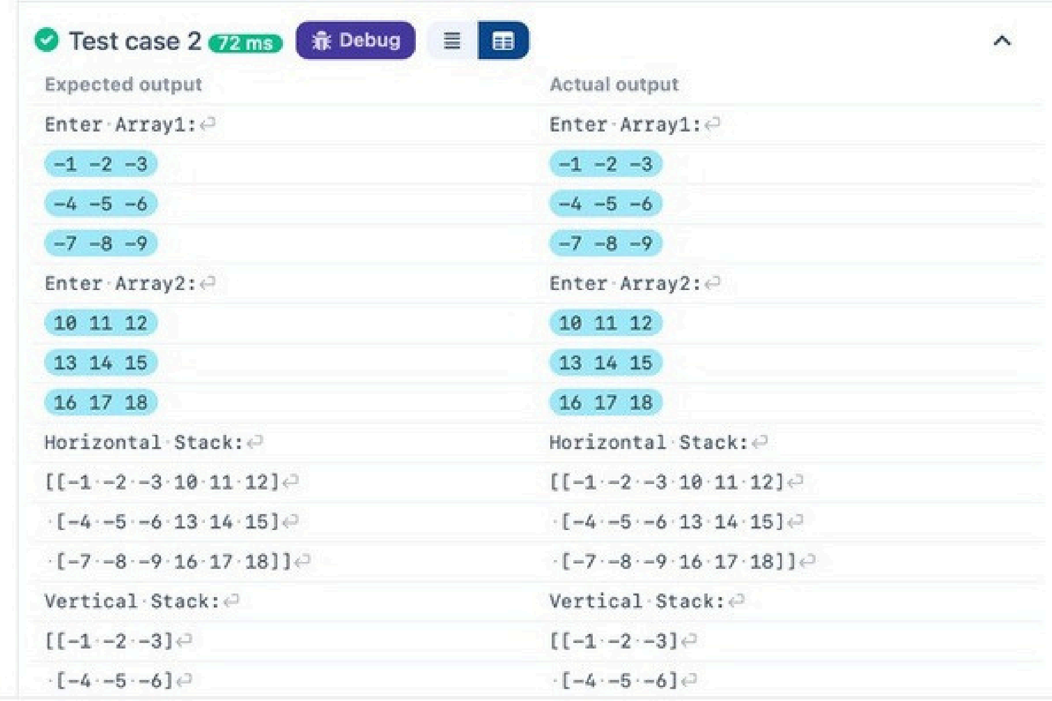
- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

INPUT

```python
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in
range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in
range(3)])

# Perform horizontal stacking (hstack)
a=np.hstack((arr1,arr2))
print("Horizontal Stack:")
print(a)


# Perform vertical stacking (vstack)
b=np.vstack((arr1,arr2))
print("Vertical Stack:")
print(b)
```

OUTPUT





3.2.3 NUMPY:CUSTOM SEQUENCE GENERATION

WriteaPython programthat takes thefollowinginputsfromtheuser:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

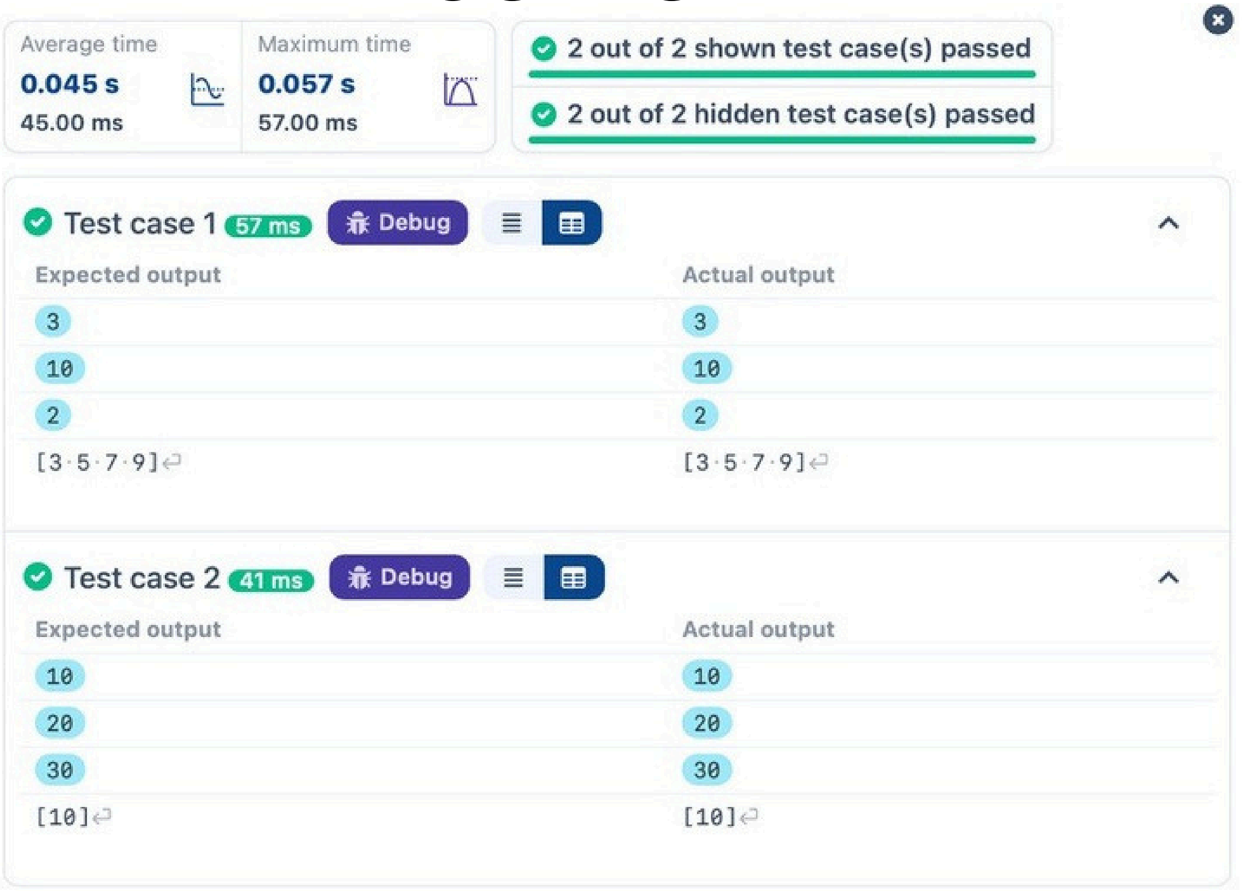
Output Format:

- The program should print the generated sequence based on the input values.

INPUT

```python
import numpy as np

# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())

# Generate the sequence using np.arange()
a=np.arange(start,stop,step)
print(a)
# Print the generated sequence
```

OUTPUT



3.2.4 NUMPY: ARITHMETIC AND STATISTICAL OPERATIONS, MATHEMATICAL OPERATIONS, BITWISE OPERATORS

You are giventwo arrays AandB.Yourtaskistocompletethefunctionarray_operations,whichwillconvertthese lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:
   - Compute the element-wise sum, difference, and product of the two arrays.
2. Statistical Operations:
   - Calculate the mean, median, and standard deviation of array A.
3. Bitwise Operations:
   - Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

INPUT

```python
import numpy as np

def array_operations(A, B):

    # Convert A and B to NumPy arrays
    A=np.array(A)
    B=np.array(B)
    # Arithmetic Operations
    sum_result = A+B
    diff_result = A-B
    prod_result = A*B

    # Statistical Operations
    mean_A = np.mean(A)
    median_A = np.median(A)
    std_dev_A = np.std(A)

    # Bitwise Operations
    and_result = A&B
    or_result = A|B
    xor_result = A^B

    # Output results with one space between each element
    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
    print("Element-wise Difference:", ' '.join(map(str,
diff_result)))
    print("Element-wise Product:", ' '.join(map(str,
prod_result)))

    print(f"Mean of A: {mean_A}")
    print(f"Median of A: {median_A}")
    print(f"Standard Deviation of A: {std_dev_A}")

    print("Bitwise AND:", ' '.join(map(str, and_result)))
    print("Bitwise OR:", ' '.join(map(str, or_result)))
    print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split()))  # Elements of array A
B = list(map(int, input().split()))  # Elements of array B
array_operations(A, B)
```

OUTPUT

Average time
0.030 s
29.67 ms

Maximum time
0.035 s
35.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 35 ms Debug

Expected output	Actual output
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32
Mean of A: 2.5	Mean of A: 2.5
Median of A: 2.5	Median of A: 2.5
Standard Deviation of A: 1.118033988749895	Standard Deviation of A: 1.118033988749895
Bitwise AND: 1 2 3 0	Bitwise AND: 1 2 3 0
Bitwise OR: 5 6 7 12	Bitwise OR: 5 6 7 12
Bitwise XOR: 4 4 4 12	Bitwise XOR: 4 4 4 12

3.2.5 NUMPY:CUSTOM SEQUENCE GENERATION

Thegivencodetakes a list ofintegersasinputandconverts itintoaNumPyarray.Yourtaskistocompletethecode by: Creating a view of the original_array and assigning it to view_array.

- Creating a copy of the original_array and assigning it to copy_array.
-

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

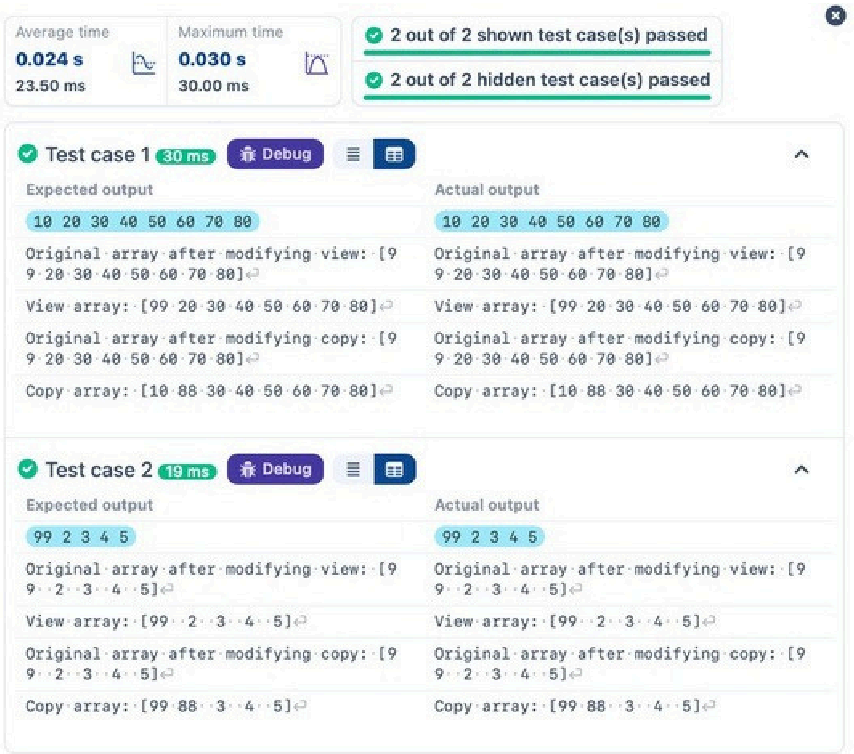
Output Format:

- After modifying the view:Original array after modifying view: **<original_array>**
- View array: **<view_array>**
- After modifying the copy: Original array after modifying copy: <**original_array**>
- Copy array: <**copy_array**>

INPUT

```python
import numpy as np

inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)

# Create a view
view_array =original_array.view()

# Create a copy
copy_array =original_array.copy()

# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)

# Modify the copy
copy_array[1] = 88
print("Original array after modifying copy:", original_array)
print("Copy array:", copy_array)
```

OUTPUT



3.2.6 NUMPY: SEARCHING, SORTING, COUNTING, BROADCASTING

Thegivencode in the editor takes a single array, array1,asspace-separatedintegersasinputfromtheuser.
Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:
1. Searching: Find the indices where search_value appears in array1 and print these indices.
2. Counting: Count how many times count_value appears in array1 and print the count.
3. Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
4. Sorting: Sort array1 in ascending order and print the sorted array.

Input Format:

1.A single line containing space-separated integers representing array1.
2.An integer search_value represents the value to search for in the array.
3.An integer count_value represents the value to count in the array.
4.An integer broadcast_value represents the value to add to each element of the array.

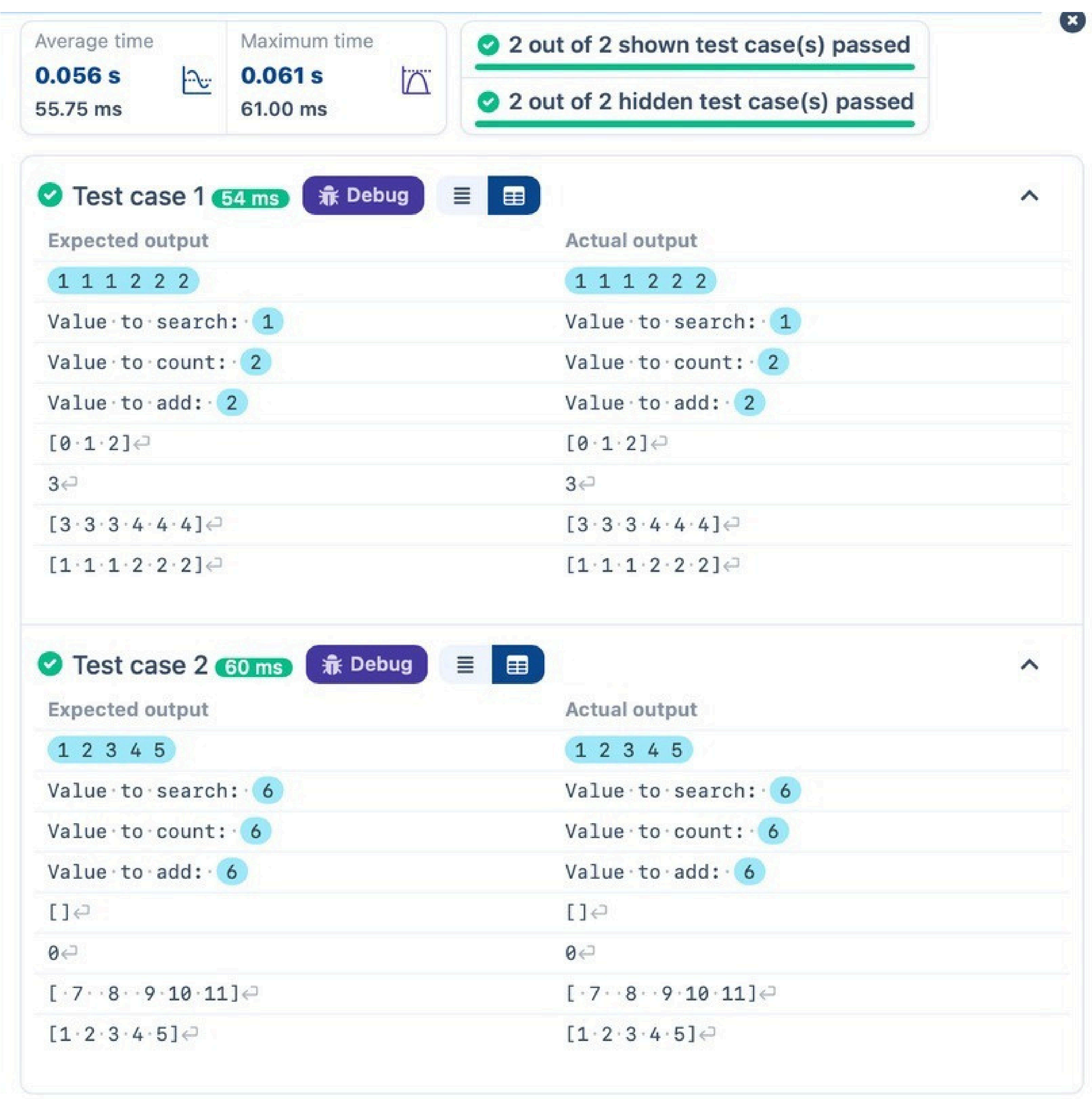
Output Format:

1.The indices where search_value occurs in array1.
2.The count of occurrences of count_value in array1.
3.The array after adding the broadcast_value to each element.
4.The sorted array.

INPUT

```
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split())))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

# Find indices where value matches in array1
a=np.where(array1==search_value)[0]
print(a)
# Count occurrences in array1
b=np.count_nonzero(array1==count_value)
print(b)
# Broadcasting addition
c=array1 + broadcast_value
print(c)
# Sort the first array
d=np.sort(array1)
print(d)
```

OUTPUT



3.2.7 STUDENT DATA ANALYSIS AND OPERATIONS

WriteaPythonprogram that takes the file name of a CSV file containingstudentdetails,includingrollnumbersandtheirmarks in three subjects as input, reads the data, and performs the following operations:

- Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.
- Find total students: Determine the total number of students in the dataset.
- Print all student roll numbers: Extract and print the roll numbers of all students.
- Print Subject 1 marks: Extract and print the marks of all students in Subject 1.
- Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.
- Find maximum marks in Subject 3: Identify the highest marks in Subject 3.
- Print all subject marks: Display the marks of all students for each subject.
- Find total marks of students: Compute the total marks for each student across all subjects.
- Find the average marks of each student: Compute the average marks for each student.
- Find average marks of each subject: Compute the average marks for all students in each subject.
- Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.
- Find average marks of Subject 1 and Subject 3: Compute the average marks for Subject 1 and Subject 3.
- Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.
- Find the roll number of the student with minimum marks in Subject 2: Identify the student with the lowest marks in Subject 2 and print their roll number.
- Find the roll number of students who scored 24 marks in Subject 2: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- Find the count of students who got less than 40 marks in Subject 1: Count the number of students who scored less than 40 marks in Subject 1.
- Find the count of students who got more than 90 marks in Subject 2: Count the number of students who scored more than 90 marks in Subject 2.
- Find the count of students who scored >=90 in each subject: Count the number of students who scored 90 or more marks in each subject.
- Find the count of subjects in which each student scored >=90: Determine how many subjects each student scored 90 or more marks in.
- Print Subject 1 marks in ascending order: Sort and print the marks of students in Subject 1 in ascending order.
- Print students who scored between 50 and 90 in Subject 1: Display students who scored marks between 50 and 90 in Subject 1.
- Find index positions of students who scored 79 in Subject 1: Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

INPUT

```python
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

# 1. Print all student details
print("All student Details:\n", a)

# 2. print total students
print("Total Students:", len(a))

# 3. Print all student Roll numbers
print("All Student Roll Nos", a[:,0])

# 4. Print subject 1 marks
print("Subject 1 Marks", a[:,1])

# 5. print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(a[:,2]))

# 6. print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(a[:,3]))

# 7. Print All subject marks
print("All subject marks:", a[:,1:])

# 8. print Total marks of students
total_marks = np.sum(a[:,1:], axis=1)
print("Total Marks", total_marks)
# 9. print average marks of each student
avg_student = np.mean(a[:,1:], axis=1)
print(np.round(avg_student, 1))

# 10. print average marks of each subject
avg_subject = np.mean(a[:,1:], axis=0)
print("Average Marks of each subject", np.round(avg_subject,
1))

# 11. print average marks of S1 and S2
avg_s1_s2 = np.mean(a[:,1:3], axis=0)
print("Average Marks of S1 and S2", np.round(avg_s1_s2, 1))

# 12. print average marks of S1 and S3
print("Average Marks of S1 and S3",
np.round([np.mean(a[:,1]), np.mean(a[:,3])], 1))

# 13. print Roll number who got maximum marks in Subject 3
print("Roll no who got maximum marks in Subject 3",
a[np.argmax(a[:,3]), 0])

# 14. print Roll number who got minimum marks in Subject 2
print("Roll no who got minimum marks in Subject 2",
a[np.argmin(a[:,2]), 0])

# 15. print Roll number who got 24 marks in Subject 2
i = np.where(a[:,2] == 24)
print("Roll no who got 24 marks in Subject 2", a[i,0])

# 16. print count of students who got marks in Subject 1 < 40
print("Count of students who got marks in Subject 1 < 40",
np.sum(a[:,1] < 40))

# 17. print count of students who got marks in Subject 2 > 90
print("Count of students who got marks in Subject 2 > 90:",
np.sum(a[:,2] > 90))

# 18. print count of students in each subject who got marks
>= 90
count_ge90 = [np.sum(a[:,1] >= 90), np.sum(a[:,2] >= 90),
np.sum(a[:,3] >= 90)]
print("Count of students in each subject who got marks >=
90:", str(count_ge90).replace(',',''))

# 19. print count of subjects in which each student got marks
>= 90
print("Roll no:", a[:,0])
print("Count of subjects in which student got marks >= 90:",
np.sum(a[:,1:] >= 90, axis=1))

# 20. Print S1 marks in ascending order
print(np.sort(a[:,1]))

# 21. Print students who scored between 50 and 90 in Subject 1
mask = (a[:,1] >= 50) & (a[:,1] <= 90)
print(a[mask])
print(a)

# 22. Print the index positions of students who scored 79 in
Subject 1
i = np.where(a[:,1] == 79)[0]
print("(array(["+", ".join(map(str, i))+"], dtype=int32),)")
```

OUTPUT